

Muhammad Usman Omer

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EDUCATION

University of British Columbia

Vancouver, BC

Bachelor of Science in Computer Science

Sep. 2023 – May 2027

- Dean's List (2024)

EXPERIENCE

Machine Learning Intern

Aug 2021 – Oct 2021

- Developed a machine learning model using R to track illegal license plates via CCTV, improving accuracy by 25% in identifying obscured plates.
- Collaborated with a team of 4 engineers to implement and test the model in a real-world setting, resulting in a 30% reduction in processing time.

Punjab Information Technology Board

Lahore, Pakistan

Machine Learning Intern

Jun 2022 – Jul 2022

Dubai, UAE

- Applied machine learning techniques and big data analytics to analyze trends in the Saudi Stock Exchange (TASI), contributing to a 12% improvement in short-term prediction accuracy.
- Enhanced prediction accuracy by 12% through the implementation of KNN algorithms and optimization of preprocessing techniques.

Himmah Capital Limited

PROJECTS

Ticket Booking Application | Java, Spring Boot, Spring Security, React, Vite

- Designed and implemented a robust ticket booking platform with secure login and registration functionality featuring a modern, responsive UI built with React and Vite .
- Utilized Spring Data JPA to streamline CRUD operations for managing user bookings, cancellations, and updates, ensuring seamless interaction with the backend database.
- Implemented Spring Security and JWT-based authentication to safeguard user data and restrict access to authorized endpoints, enhancing overall application security.
- Developed RESTful API endpoints using Spring Boot REST controllers , enabling efficient communication between the frontend and backend.
- Introduced a well-structured service layer to handle business logic, ensuring scalable and maintainable separation of concerns within the application.
- Backend Repository | Frontend Repository | Live Application
- FUT Team Builder Application | Java, Swing, JUnit, Mockito
- Developed a comprehensive application for FIFA Ultimate Team (FUT) enabling users to create, manage, and optimize custom teams with an interactive gaming-style interface using Java Swing.
- Implemented secure login and registration functionality using Java's Spring Security.

access control for users and admins.

- Designed a complex service layer with algorithms to calculate ke rating , and total price , improving the user experience.
- Integrated a community feature that allows users to save an others to browse, filter, and like popular team builds.
- Achieved 90% code coverage through comprehensive cases for all service layer components and user interactions
 - Source Code
SFTWR Labs Web Application | Java, Spring Boot, Next.js
 - Designed and developed a full-stack business with Spring Boot and a dynamic frontend using
 - Implemented a RESTful API architecture through Spring REST endpoints.
 - Developed an Admin Panel with Spring Mail for email communications:
 - Visit SFTWR Labs | Source Code

TECHNICAL SKILLS

Languages : Java, Python, SQL, JavaScript, HTML/ CSS

Frameworks : Spring Boot, React, Hibernate

Tools : Git, Redis Cache, Amazon Bedrock, Open AI API, Apache Kafka, AWS, Docker, Kubernetes